Global Earthquake Analysis

Complete Python Source Code

Includes the data-cleaning script, analysis script, and all code cells from the Google Colab notebook. Code is reproduced from your project files, with page breaks and line wrapping for readability.

clean_earthquake_data.py

```
import pandas as pd
# Load the original dataset
df = pd.read_csv('global_earthquakes_seismic.csv')
print('Original shape:', df.shape)
# Replace blank text values with missing values
text_cols = df.select_dtypes(include='object').columns
df[text_cols] = df[text_cols].replace(r'^\s*$', pd.NA, regex=True)
# Remove duplicate event IDs and duplicate rows
df = df.drop_duplicates().drop_duplicates(subset='id')
# Keep only earthquakes (the source includes two landslides)
df = df.loc[df['type'].eq('earthquake')].copy()
# Convert Unix epoch milliseconds to UTC datetime
df['time'] = pd.to_datetime(df['time'], unit='ms', utc=True, errors='coerce')
# Remove records with missing critical fields or invalid physical values
df = df.dropna(subset=['id', 'time', 'latitude', 'longitude', 'depth', 'mag'])
df = df.loc[
    df['latitude'].between(-90, 90)
    &amp;amp; df['longitude'].between(-180, 180)
    &amp;amp; df['depth'].ge(0)
    &amp;amp; df['mag'].ge(0)
].copy()
# Add fields useful for grouping and visualization
df['year'] = df['time'].dt.year
df['month'] = df['time'].dt.strftime('%Y-%m')
df['date'] = df['time'].dt.strftime('%Y-%m-%d')
# Remove redundant narrative fields, retaining original event and category fields
df = df.drop(columns=['evidence_span', 'cot_rationale', 'title'], errors='ignore')
# Sort events chronologically
df = df.sort_values('time').reset_index(drop=True)
df.to_csv('global_earthquakes_cleaned.csv', index=False)
print('Cleaned shape:', df.shape)
print('Remaining missing values:', int(df.isna().sum().sum()))
print('Duplicate IDs:', int(df['id'].duplicated().sum()))
print('Saved: global_earthquakes_cleaned.csv')
```

earthquake_analysis.py

```
"""Earthquake analysis: manipulation, charts and descriptive statistics."""
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
ROOT = Path(__file__).resolve().parent
OUT = ROOT / 'visualizations'
OUT.mkdir(exist_ok=True)
sns.set_theme(style='whitegrid')
df = pd.read_csv(ROOT / 'global_earthquakes_cleaned.csv')
df['time'] = pd.to_datetime(df['time'], utc=True, format='mixed')
# Sorting: find the strongest recorded earthquakes.
print('Top ten earthquakes by magnitude:')
print(df.sort_values('mag', ascending=False)[['place','mag','depth']].head(10).to_string(index=Fals
    e))
# Filtering: focus on earthquakes of magnitude 6.0 or greater.
strong = df.loc[df['mag'] &gt;= 6.0]
print('Magnitude &gt;= 6:', len(strong), 'of', len(df), '(',round(100*len(strong)/len(df),2),'%)')
# Grouping: compare event counts, magnitude and depth by tectonic zone.
region = 'tectonic_boundary_zone'
summary = df.groupby(region).agg(earthquake_count=('id','count'), average_magnitude=('mag','mean'),
     average_depth_km=('depth','mean')).sort_values('earthquake_count',ascending=False)
print('By tectonic zone:\n', summary.round(2))
# Descriptive statistics and correlation describe central tendency, variation, and association.
print('Magnitude and depth statistics:\n',df[['mag','depth']].describe().round(3))
print('Magnitude mode:',df['mag'].mode().iloc[0])
print('Magnitude skewness:',round(df['mag'].skew(),3))
print('Pearson correlation (magnitude/depth):',round(df['mag'].corr(df['depth']),3))
fig, ax = plt.subplots(figsize=(11,6))
counts = df[region].value_counts()
sns.barplot(x=counts.values,y=counts.index,color='steelblue',ax=ax)
ax.set(title='Earthquakes by Tectonic Region',xlabel='Number of Earthquakes',ylabel='Tectonic Regio
    n')
fig.tight_layout();fig.savefig(OUT/'earthquakes_by_region.png',dpi=160);plt.close(fig)
fig,ax=plt.subplots(figsize=(10,6))
sns.histplot(data=df,x='mag',bins=25,color='orange',edgecolor='black',ax=ax)
ax.set(title='Distribution of Earthquake Magnitudes',xlabel='Magnitude',ylabel='Number of Earthquak
    es')
fig.tight_layout();fig.savefig(OUT/'magnitude_distribution.png',dpi=160);plt.close(fig)
fig,ax=plt.subplots(figsize=(10,6))
sns.scatterplot(data=df,x='depth',y='mag',alpha=.25,color='purple',label='Earthquakes',ax=ax,s=18)
ax.set(title='Earthquake Magnitude vs Depth',xlabel='Depth (km)',ylabel='Magnitude');ax.legend()
fig.tight_layout();fig.savefig(OUT/'magnitude_vs_depth.png',dpi=160);plt.close(fig)
monthly=df.set_index('time').resample('MS').size()
fig,ax=plt.subplots(figsize=(12,6))
ax.plot(monthly.index,monthly.values,marker='o',color='green',label='Monthly earthquakes')
ax.set(title='Recorded Earthquakes by Month',xlabel='Month',ylabel='Number of Earthquakes')
ax.tick_params(axis='x',rotation=45);ax.legend()
fig.tight_layout();fig.savefig(OUT/'earthquakes_over_time.png',dpi=160);plt.close(fig)
fig,ax=plt.subplots(figsize=(7,5))
sns.heatmap(df[['mag','depth']].corr(),annot=True,cmap='coolwarm',vmin=-1,vmax=1,ax=ax)
ax.set_title('Magnitude and Depth Correlation')
fig.tight_layout();fig.savefig(OUT/'correlation_heatmap.png',dpi=160);plt.close(fig)
print('Saved five charts to:',OUT)
```

Colab notebook — code cell 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from pathlib import Path
try:
    from google.colab import files
    uploaded = files.upload()
    source_file = next(iter(uploaded))
except ImportError:
    source_file = 'global_earthquakes_seismic.csv'
df = pd.read_csv(source_file)
print('Original shape:', df.shape)
display(df.head())
print('Missing values:', df.isna().sum().sum())
print('Duplicate IDs:', df.id.duplicated().sum())
print('Event types:',df.type.value_counts())
```

Colab notebook — code cell 4

```
df = df.replace(r'^\s*$', np.nan, regex=True)
df = df.drop_duplicates().drop_duplicates(subset='id')
df = df[df['type'].eq('earthquake')].copy()
df['time'] = pd.to_datetime(df['time'],unit='ms',utc=True,errors='coerce')
for col in ['latitude','longitude','depth','mag']:
    df[col] = pd.to_numeric(df[col],errors='coerce')
df = df.dropna(subset=['id','time','latitude','longitude','depth','mag'])
df = df[df.latitude.between(-90,90) &amp; df.longitude.between(-180,180) &amp; (df.depth &gt;= 0) &amp; (df.mag &gt;=
     0)].copy()
df['year'] = df.time.dt.year
df['month'] = df.time.dt.strftime('%Y-%m')
df['date'] = df.time.dt.strftime('%Y-%m-%d')
df = df.drop(columns=['evidence_span','cot_rationale','title'],errors='ignore')
df = df.sort_values('time').reset_index(drop=True)
df.to_csv('global_earthquakes_cleaned.csv',index=False)
print('Cleaned shape:',df.shape)
```

Colab notebook — code cell 6

```
display(df.sort_values('mag',ascending=False)[['place','mag','depth']].head(10))
strong = df[df.mag &gt;= 6.0]
print('Magnitude 6.0+ events:',len(strong))
region='tectonic_boundary_zone'
summary = df.groupby(region).agg(count=('id','count'),mean_mag=('mag','mean'),mean_depth=('depth','
    mean')).sort_values('count',ascending=False)
display(summary.round(2))
```

Colab notebook — code cell 8

```
display(df[['mag','depth']].describe())
print('Magnitude mode:',df.mag.mode().iloc[0])
print('Magnitude skewness:',round(df.mag.skew(),3))
print('Magnitude-depth correlation:',round(df.mag.corr(df.depth),3))
print('Percentage magnitude 6+:',round(100*len(strong)/len(df),2))
```

Colab notebook — code cell 10

```
sns.set_theme(style='whitegrid')
Path('visualizations').mkdir(exist_ok=True)
counts=df[region].value_counts()
fig,ax=plt.subplots(figsize=(11,6))
sns.barplot(x=counts.values,y=counts.index,color='steelblue',ax=ax)
ax.set(title='Earthquakes by Tectonic Region',xlabel='Number of Earthquakes',ylabel='Tectonic Regio
    n')
fig.tight_layout();fig.savefig('visualizations/earthquakes_by_region.png',dpi=160);plt.show()
```

Colab notebook — code cell 11

```
fig,ax=plt.subplots(figsize=(10,6))
sns.histplot(data=df,x='mag',bins=25,color='orange',edgecolor='black',ax=ax)
ax.set(title='Distribution of Earthquake Magnitudes',xlabel='Magnitude',ylabel='Number of Earthquak
    es')
fig.tight_layout();fig.savefig('visualizations/magnitude_distribution.png',dpi=160);plt.show()
```

Colab notebook — code cell 12

```
fig,ax=plt.subplots(figsize=(10,6))
sns.scatterplot(data=df,x='depth',y='mag',alpha=.25,color='purple',label='Earthquakes',ax=ax,s=18)
ax.set(title='Earthquake Magnitude vs Depth',xlabel='Depth (km)',ylabel='Magnitude');ax.legend()
fig.tight_layout();fig.savefig('visualizations/magnitude_vs_depth.png',dpi=160);plt.show()
```

Colab notebook — code cell 13

```
monthly=df.set_index('time').resample('MS').size()
fig,ax=plt.subplots(figsize=(12,6))
ax.plot(monthly.index,monthly.values,marker='o',color='green',label='Monthly earthquakes')
ax.set(title='Recorded Earthquakes by Month',xlabel='Month',ylabel='Number of Earthquakes')
ax.tick_params(axis='x',rotation=45);ax.legend()
fig.tight_layout();fig.savefig('visualizations/earthquakes_over_time.png',dpi=160);plt.show()
```

Colab notebook — code cell 14

```
fig,ax=plt.subplots(figsize=(7,5))
sns.heatmap(df[['mag','depth']].corr(),annot=True,cmap='coolwarm',vmin=-1,vmax=1,ax=ax)
ax.set_title('Magnitude and Depth Correlation')
fig.tight_layout();fig.savefig('visualizations/correlation_heatmap.png',dpi=160);plt.show()
```

requirements.txt

```
pandas
numpy
matplotlib
seaborn
```